

AELL101

Lecture Notes

Group 5

25th March - 3rd April

1. Semiconductors

Conductors - Large number of free electrons

Insulator - No free electrons

Semiconductor - Some free electrons

Semiconductors are materials whose electrical conductivity lies between that of conductors and insulators. Their conductivity can be modified by doping and external conditions like temperature or electric fields, making them essential for electronic devices.

Table 1: Variation of conductivity with different materials

Material	Type	Conductivity [S/m]	Valence Electrons
Copper	Conductor	5×10^7	1
Silver	Conductor	6×10^7	1
Silicon	Semiconductor	10^{-3}	4
Germanium	Semiconductor	2.2	4
Glass	Insulator	10^{-14} to 10^{-15}	-
Sulphur	Insulator	10^{-15}	6

The valence band is the energy band containing electrons bound to atoms; these electrons do not contribute to conduction unless they gain energy.

The conduction band is the energy band where free electrons can move and conduct electricity after gaining enough energy to jump from the valence band.

In conductors, the conduction band and valence band overlap, so there is no band gap, allowing free movement of electrons.

In insulators, the band gap is large (greater than 5 eV), so electrons cannot easily move to the conduction band, preventing conduction.

Semiconductors have small band gap (typically less than 3 eV) and require external energy to push electrons into the conduction band. Number of free electrons is very less, so it causes weaker current.

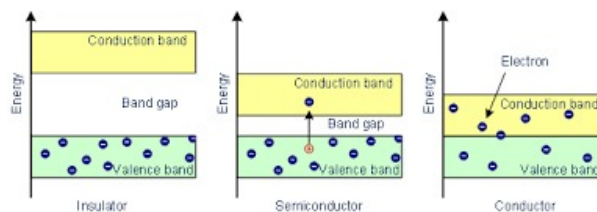


Figure 1: Diagram showing the variation of the band gap for different materials

Semiconductors, at lower temperature, act as insulators and at higher temperatures act as conductors.

2. P type and N type semiconductors

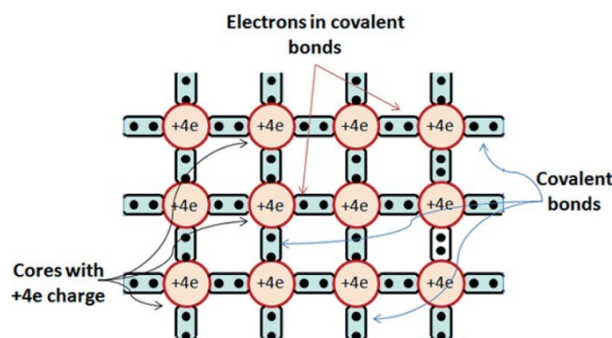


Figure 2: Diagram showing sharing of electrons between atoms of silicon

We dope semiconductors with group 3 and 5 elements:

2.1 P-type semiconductors

Aluminum, a group 3 element, is trivalent and creates a **P-type semiconductor** when it replaces a silicon atom in the crystal. Since aluminum has only three valence electrons, it forms three covalent bonds, leaving one bond incomplete. This missing electron creates a **hole**, which acts as a positive charge carrier and contributes to electrical conduction in the material.

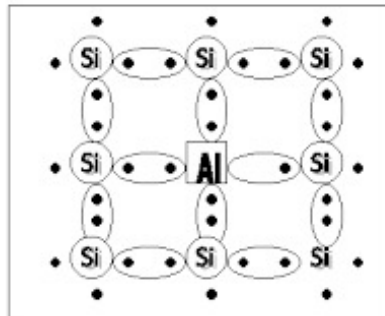


Figure 3: Diagram showing the creation of holes when aluminium atoms are doped in silicon

2.2 N-type semiconductors

When phosphorus, a group 5 element, is doped into silicon, it brings **five valence electrons**. Four are used to form covalent bonds with neighboring silicon atoms, leaving one extra electron free. This free electron becomes a **mobile charge carrier**, making the material an **N-type semiconductor**.

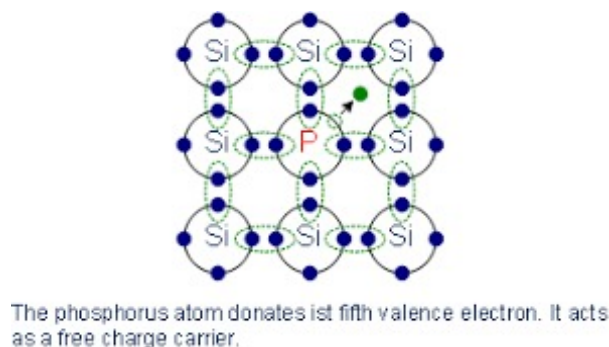


Figure 4: Diagram showing the the free electrons emerging as a result of doping phosphorous in silicon

3. Diodes

A diode is a fundamental electronic component that allows current to flow in only one direction. It is created by forming a **P-N junction**, which is the boundary between two types of semiconductor materials:

- **P-type (positive-type):** Contains an abundance of *holes* (missing electrons), created by doping with elements such as boron.
- **N-type (negative-type):** Contains a surplus of *free electrons*, introduced by doping with elements like phosphorus.

When these materials are joined, electrons from the N-side diffuse into the P-side and recombine with holes. This process leads to the formation of a **depletion region** — a zone near the junction where no free charge carriers exist. As a result, an internal **electric field** develops across this region, which acts as a potential barrier and resists further movement of charge carriers.

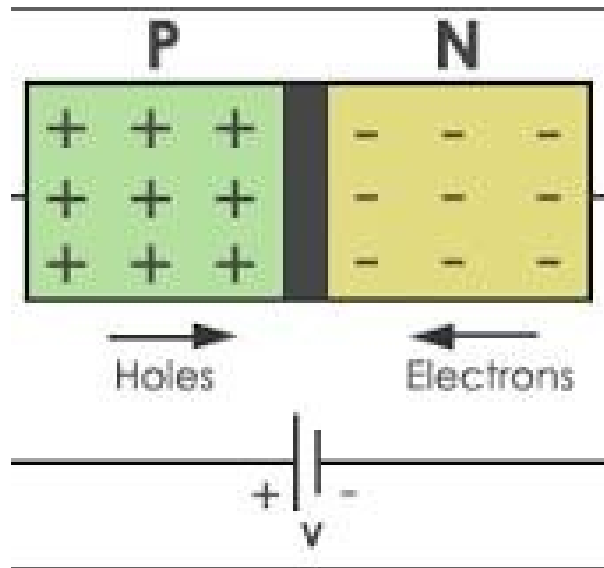


Figure 5: Diagram showing the p-n junction

Current can only flow through the diode when it is forward biased, meaning the P-side is connected to the positive terminal of a power source and the N-side to the negative. This reduces the width of the depletion region and allows charge carriers to move across the junction.

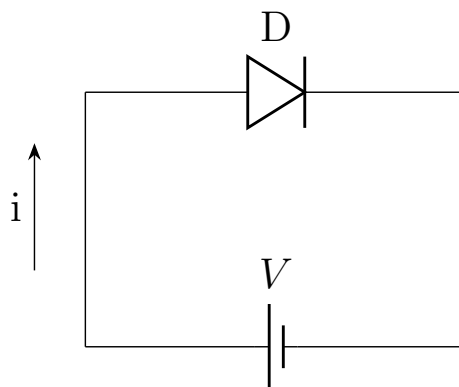


Figure 6: Circuit showing a forward bias diode

In reverse bias, where the P-side is connected to the negative terminal and the N-side to the positive, the depletion region widens and current is blocked.

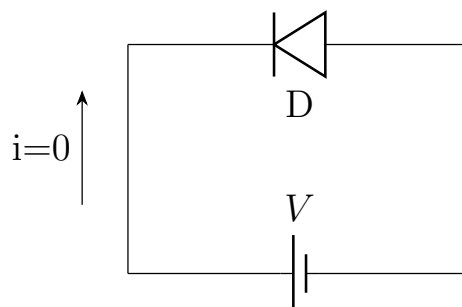


Figure 7: Circuit showing a reverse bias diode

3.1 Ideal and non-ideal diodes

An **ideal diode** is a theoretical model that assumes perfect behavior. It is used for simplifying circuit analysis and assumes the following:

- In **forward bias**, it conducts perfectly (like a closed switch). As soon as any positive voltage is applied, it allows unlimited current with zero voltage drop.
- In **reverse bias**, it blocks all current completely (like an open switch), no matter how much reverse voltage is applied.

Think of an ideal diode as a perfect on/off switch: a perfect conductor in one direction, and a perfect insulator in the other.

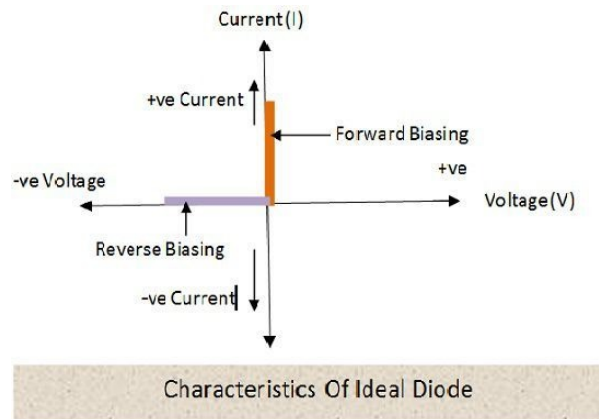


Figure 8: Diagram showing the characteristics of an ideal diode

A **non-ideal diode**, or real diode, behaves similarly but with practical limitations:

- In **forward bias**, it requires a small threshold voltage (also called the cut-in, knee or forward bias voltage) to start conducting:
 - About 0.7V for silicon diodes.
 - About 0.3V for germanium diodes.
- In **reverse bias**, it blocks most current, but a small leakage current still flows. If the reverse voltage exceeds a certain limit (called breakdown voltage), the diode allows large reverse current, which can damage the diode or be used intentionally in special diodes like Zener diodes.

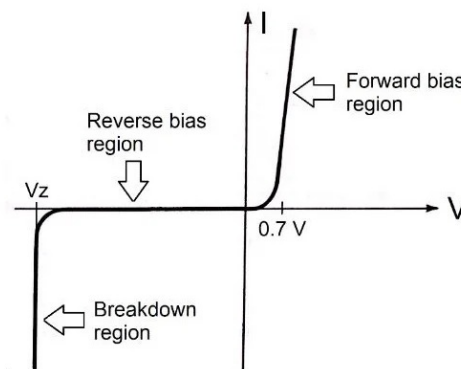
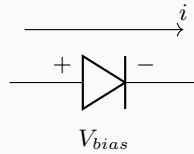


Figure 9: Diagram showing characteristics of a non ideal diode

How to solve Diode circuits?

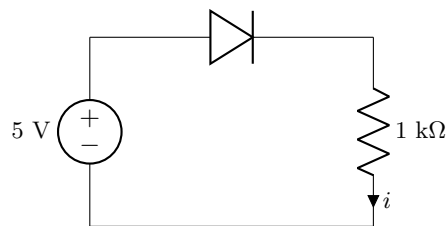
Diode Behavior

Consider a diode with forward bias voltage V_{bias} with voltage across it being V_D and current i :



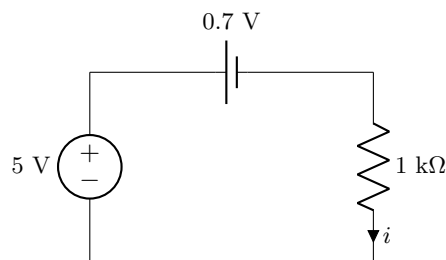
$$\left\{ \begin{array}{ll} \text{---} \overset{V_{bias}}{\text{---}} & V_D \geq V_{bias} \Rightarrow i \geq 0, \quad \text{Diode is } \mathbf{ON} \text{ (Forward Biased)} \\ \text{---} \bigcirc \quad \bigcirc \text{---} & V_D < V_{bias} \Rightarrow i = 0, \quad \text{Diode is } \mathbf{OFF} \text{ (Forward Biased, Not Conducting)} \\ \text{---} \bigcirc \quad \bigcirc \text{---} & V_D < 0, V_{bias} \Rightarrow i = 0, \quad \text{Diode is } \mathbf{OFF} \text{ (Reverse Biased)} \end{array} \right.$$

Example 1. Find the current i flowing through the following circuit consisting of a Si diode:



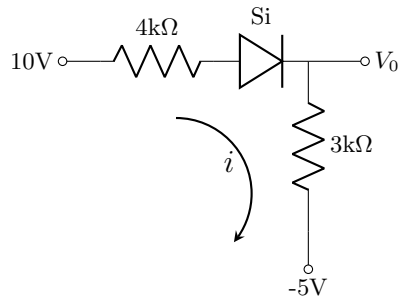
Solution:

The diode is forward biased, so we can redraw the circuit as below:



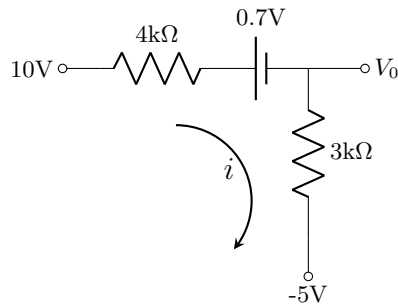
$$i = \frac{5 - 0.7}{1k} = 4.3mA$$

Example 2. Find the current i flowing and V_0 from the following circuit.



Solution:

Assume the diode is ON, then we can redraw the circuit diagram as:



Applying KVL,

$$10 - 4000i - 0.7 - 3000i = -5$$

$$i = 2.04 \text{ mA}$$

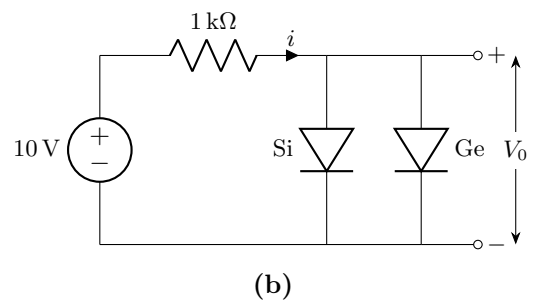
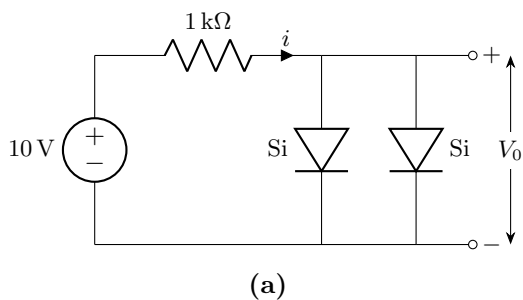
The value of i is positive, which means the diode is forward biased, and our assumption is correct.

$$V_0 - (-5) = 3000i$$

$$V_0 = 6.12 - 5$$

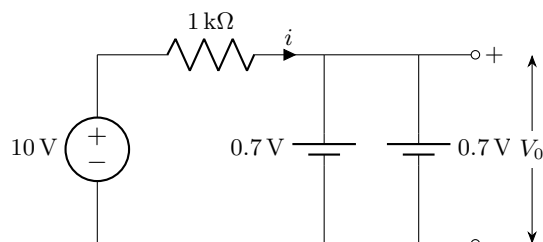
$$V_0 = 1.12 \text{ V}$$

Example 3. Find the current flowing i and the voltage V_0 across the diodes in the following circuits:



Solution: a) Two Si diodes in parallel:

Assume both the diodes are ON:



Applying KVL,

$$10 - 1000i - 0.7 = 0$$

$$i = 9.3 \text{ mA}$$

b) A Si diode and a Ge diode in parallel:

Case 1: Assume both the diodes are ON:

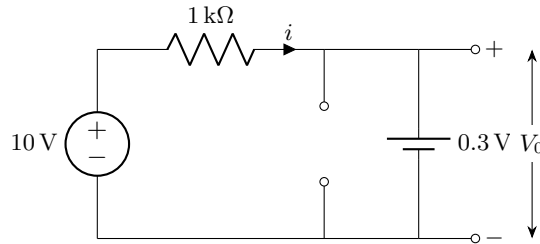
The turn-on voltages for the Si diode and Ge diode are $V_{Si} = 0.7 \text{ V}$ and $V_{Ge} = 0.3 \text{ V}$, respectively. Since both diodes are connected in parallel, they must have the same voltage across them. However, assuming both diodes are ON leads to a contradiction, as it would require two different voltages across the same nodes. Hence, this case is not possible.

Case 2: D_{Ge} is OFF and D_{Si} is ON:

If the Si diode is ON, the voltage across it must be $V_{Si} = 0.7 \text{ V}$. Since the Ge diode is OFF, the voltage across it must be less than its turn on voltage (0.3 V). However, because both diodes are connected in parallel, they must have the same voltage across them, leading to a contradiction. Therefore, this case is not possible.

Case 3: D_{Si} is OFF and D_{Ge} is ON:

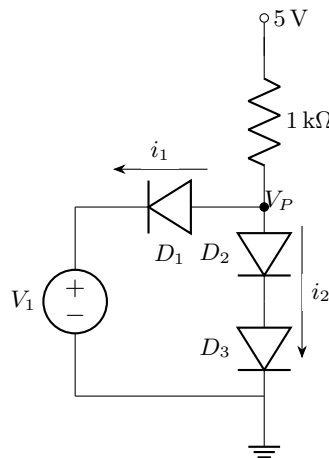
If the Ge diode is ON, the voltage across it must be $V_{Ge} = 0.3 \text{ V}$. Since both diodes are connected in parallel, the Si diode must also have the same voltage across it. However, this voltage is less than the turn on voltage of the Si diode (0.7 V), meaning the Si diode remains OFF. Since this does not lead to any contradictions, this case is possible.



$$V_0 = 0.3 \text{ V}$$

$$i = \frac{10 - 0.3}{1000} = 9.7 \text{ mA}$$

Example 4. Find the values of V_1 for which $i_2 > 0$, if all the diodes are Si diodes.



Solution:

For i_2 to flow, both diodes D_2 and D_3 are ON

$$\Rightarrow V_P = 1.4 \text{ V}$$

$$\Rightarrow V_1 = 0.7 \text{ V for } i_1 \text{ to flow.}$$

If $0 < V_1 < 0.7$ and D_1 is ON

$$\Rightarrow V_P = V_1 + 0.7$$

$$\Rightarrow V_P < 1.4$$

$$\Rightarrow D_2 \text{ is OFF}$$

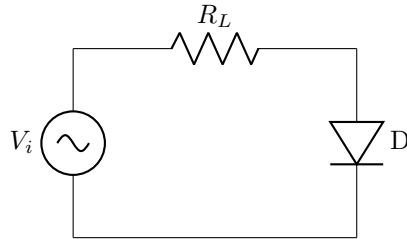
If $V_1 > 0.7$, then D_1 is **OFF** and D_2 is **ON**.

$$\Rightarrow V_P = 1.4 \text{ V and } i_1 = 0$$

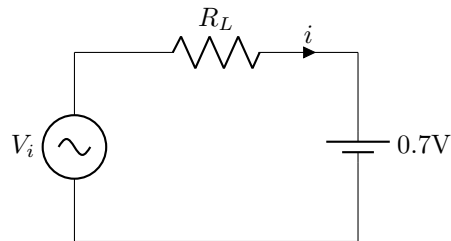
$$\Rightarrow i_2 = \frac{5-1.4}{1000} = 3.6 \text{ mA}$$

A very slight change in V_1 can switch i_2 from 0 to 3.5 mA

Example 5. Analyze the circuit given below. Predict the condition for D to be ON and plot the current vs time graph. $V_i = V_m \sin \omega t$, $V_m = 10 \text{ V}$



Solution: If D is ON,



The diode conducts when:

$$i > 0$$

$$\Rightarrow \frac{V_i - 0.7}{R_L} > 0$$

$$\Rightarrow V_i - 0.7 > 0$$

$$\Rightarrow V_i > 0.7$$

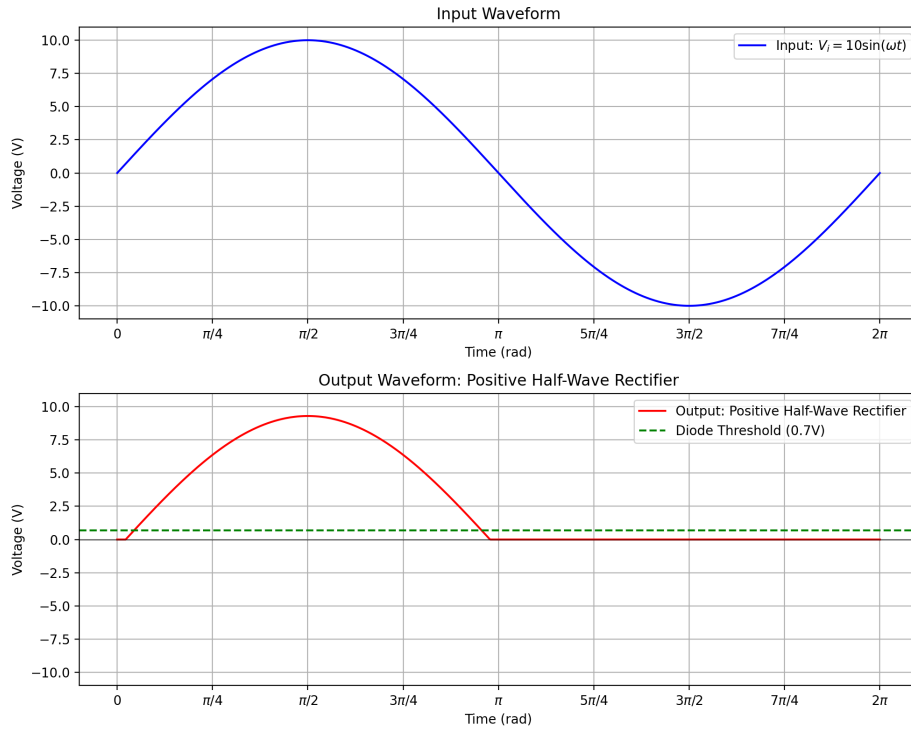
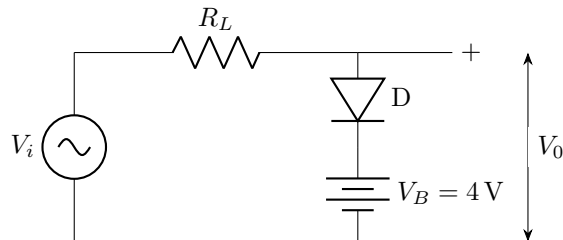


Figure 10: Input and Output waveforms for the above half wave rectifier circuit.

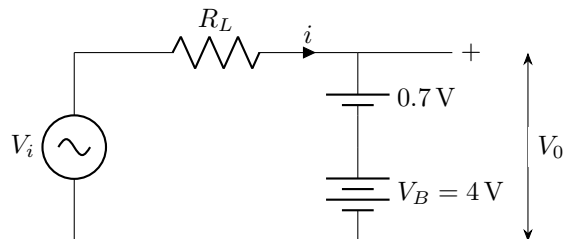
Clipper Circuits

Example 6. Find the condition for diode D to be ON in the following circuit and sketch the V_0 vs. t graph of the output signal. Given: $V_i = 10 \sin \omega t$



Solution:

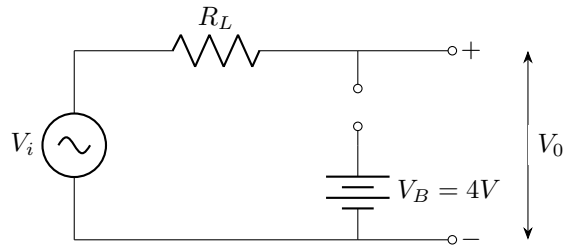
When D is ON:



The diode conducts when:

$$i > 0 \Rightarrow \frac{V_i - V_B - 0.7}{R_L} > 0 \Rightarrow V_i > V_B + 0.7 = 4.7 \text{ V}$$

When D is OFF:



Since the diode is OFF, no current flows:

$$i = 0 \Rightarrow V_0 = V_i$$

Output expression:

$$V_0 = \begin{cases} 4.7 \text{ V}, & \text{if } V_i > 4.7 \text{ V} \\ V_i, & \text{if } V_i \leq 4.7 \text{ V} \end{cases}$$

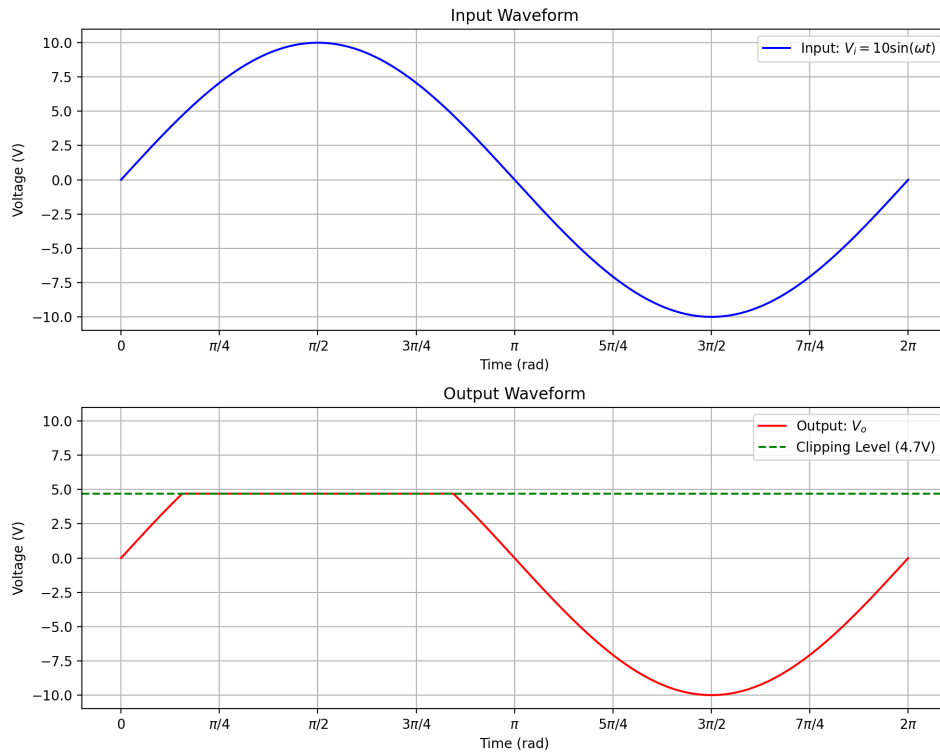
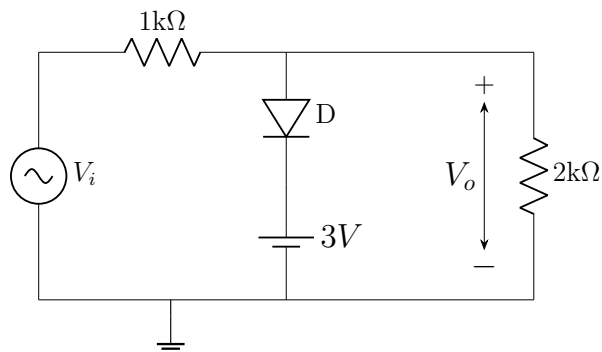


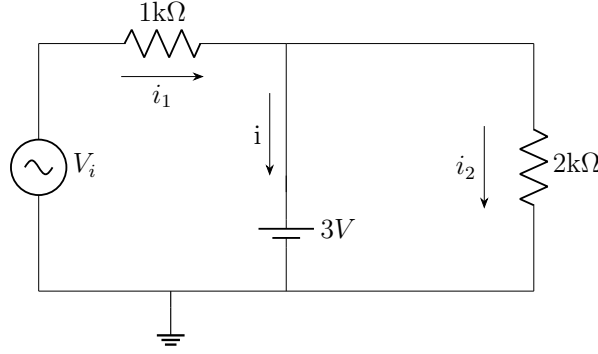
Figure 11: Input and Output waveforms for the above positive clipper circuit.

Example 7. Find the condition for diode D to be ON in the following circuit.



Solution:

When D is ON:



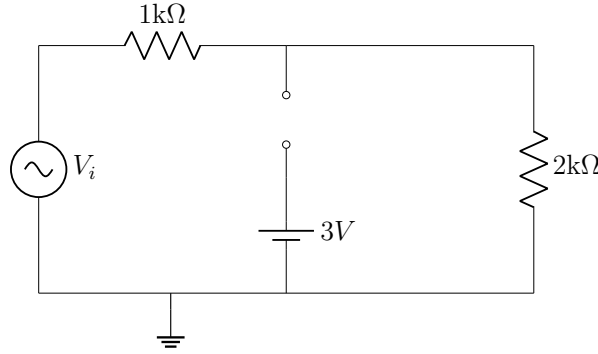
The diode conducts when:

$$i_2 = \frac{3}{2000} = 1.5 \text{ mA}, i_1 = \frac{V_i - 3}{1000}$$

$$i = i_1 - i_2 = \frac{V_i - 4.5}{1000} \text{ A}$$

$$i > 0 \quad \text{if } V_i > 4.5$$

When D is OFF:



Since the diode is OFF, no current flows through the diode:

$$V_i < 4.5 \rightarrow \text{D is OFF}, V_o = V_i \times 2000/3000 = \frac{2}{3} V_i$$

Output expression:

$$V_o = \begin{cases} 3 \text{ V}, & \text{if } V_i > 4.5 \text{ V} \\ 2/3 V_i, & \text{if } V_i \leq 4.5 \text{ V} \end{cases}$$

Half-Wave and Full-Wave Rectifier

A half wave rectifier is an electronic circuit that allows only one half (usually the positive half) of an AC input signal to pass through, while blocking the other half. It uses a single diode to convert AC into a pulsating DC voltage, making it simple but less efficient than a full wave rectifier.

A full wave rectifier is an electronic circuit that converts both halves of an AC input signal into a pulsating DC output, using either two diodes with a center-tapped transformer or four diodes in a bridge configuration.

$$\text{Average voltage across } R_L = \begin{cases} \frac{V_o}{\pi}, & \text{Half Wave Rectifier} \\ \frac{2V_o}{\pi}, & \text{Full Wave Rectifier} \end{cases}$$

Example 8. Find i and i_{avg} of the following half-rectifier circuit.

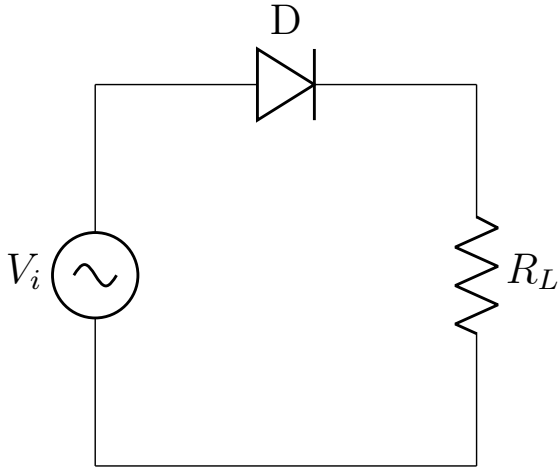


Figure 12: (a) Half-Wave Rectifier

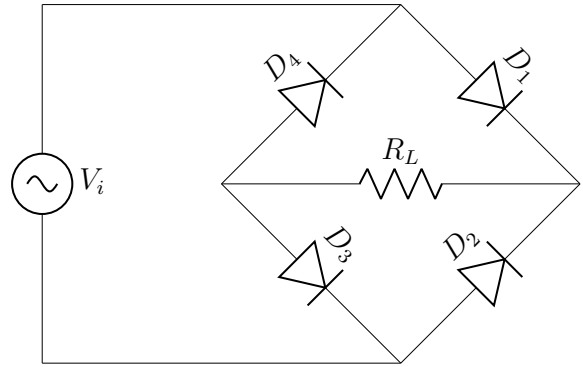


Figure 13: (b) Full-Wave Rectifier

Figure 14: Comparison of Half-Wave and Full-Wave Rectifier Circuits

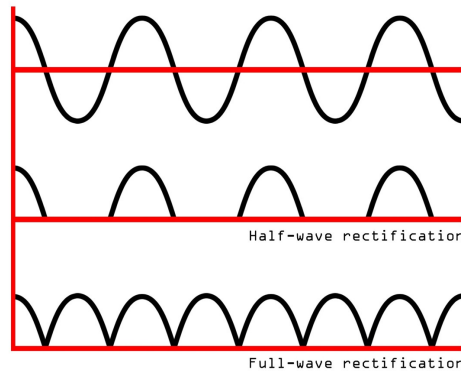
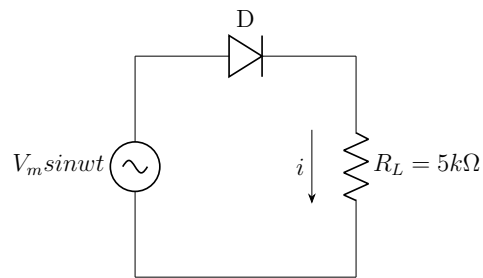


Figure 15: Half-Wave and Full-Wave Rectifier Waveforms



Solution:

$$i = \frac{V_m \times \sin \omega t}{R_L} = \frac{10 \times \sin \omega t}{5000} = 2 \sin \omega t \text{ mA}$$

$$i_{avg} = \frac{I_m}{\pi} = \frac{2}{\pi} \text{ mA}$$

Forward Biased Diode and Capacitor

When an ideal diode and capacitor are connected in series to an alternating voltage source as shown in the below circuit diagram, the diode is forward biased during the initial positive half cycle of the input from 0 to V_m , and hence charges the capacitor to V_m . As the input starts decreasing from V_m , the diode becomes reverse biased since the capacitor is at a higher voltage. And since the input never exceeds V_m , the diode remains reversed biased and the voltage across capacitor stays at V_m .

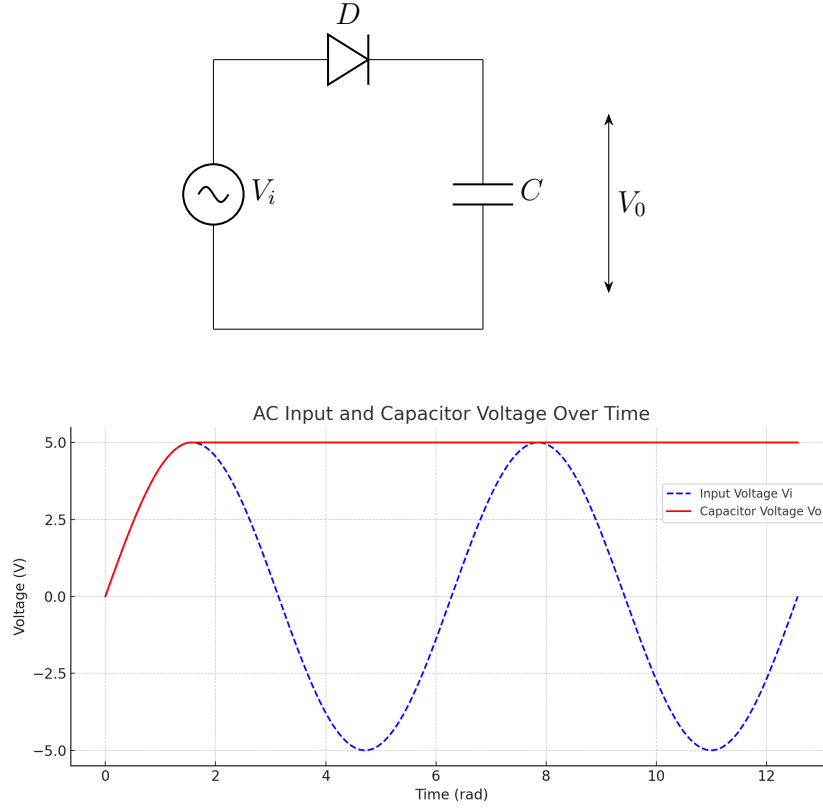


Figure 16: Input waveform as $5\sin(\omega t)$ and output waveform across the capacitor for the above circuit

When there is a resistor or load across the capacitor (in parallel), during the initial positive half-cycle from 0 to V_m , the diode is forward biased and charges the capacitor to V_m . As the input starts decreasing from V_m , the diode becomes reverse biased since the capacitor is at a higher voltage. Consequently, the capacitor begins to discharge through the load resistance, and the voltage across the capacitor (which is equal to the voltage across the load, as they are in parallel) decreases exponentially ($V = V_m e^{-t/\tau}$). When the input voltage exceeds the capacitor voltage again, the diode becomes forward biased, and the capacitor starts charging once more. This cycle then repeats. The output waveform across the capacitor and load is shown in figure 17.

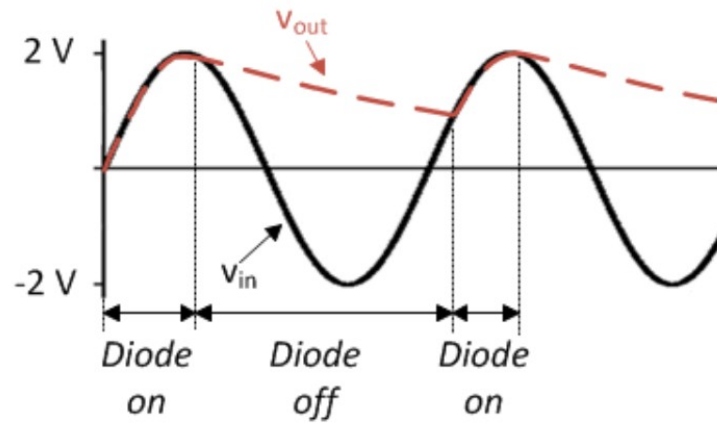


Figure 17: Output Waveform when there is a Resistor in Parallel to Capacitor and the Input is $2\sin(\omega t)$

Contribution

Ehsan- 2nd April lecture notes and formatting it in overleaf

Farhan- 3rd April lecture notes and formatting it in overleaf

Himanshu - 25th March lecture notes and formatting it in overleaf

Abhiram - Drawing circuits and graphs and overall supervision on the notes